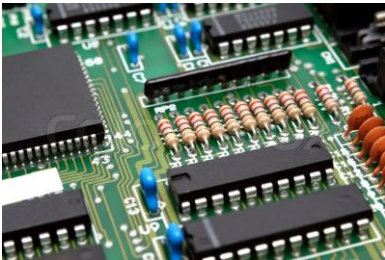
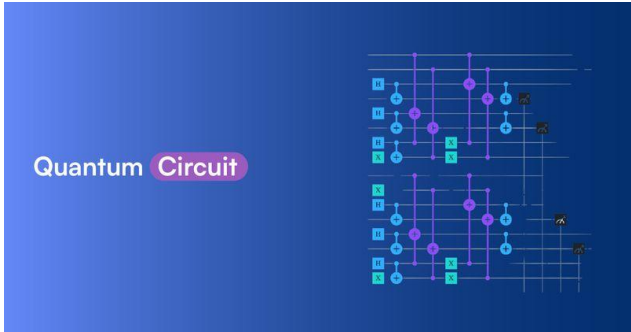
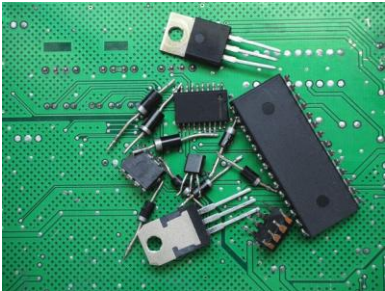
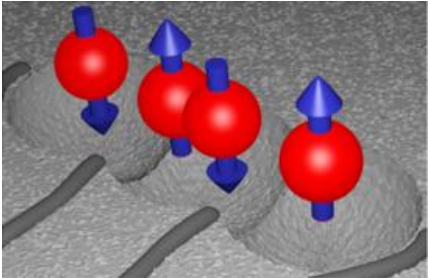


Introduction to Quantum Computing (L-2)

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Scope of this Course

L4:	Number, Text, Video, File, etc.	Number, Simulation Results, Optimization results, etc.
L3:	0 1 0 0 1 0..... (Boolean Algebra)	$0\rangle 1\rangle 0\rangle 0\rangle 1\rangle 0\rangle.....$ (Linear Algebra)
L2:		
L1:		
L0:	Circuit Theory, EM Theory, Semiconductor Devices	Quantum Theory, Spin Physics, Quantum Devices

Scope of this Course

What This Course Focuses On

- We will study quantum circuits, but not the hardware electronics behind them.
- Our focus is on quantum logic, linear algebra, and algorithms.
- We will program and simulate circuits using Qiskit

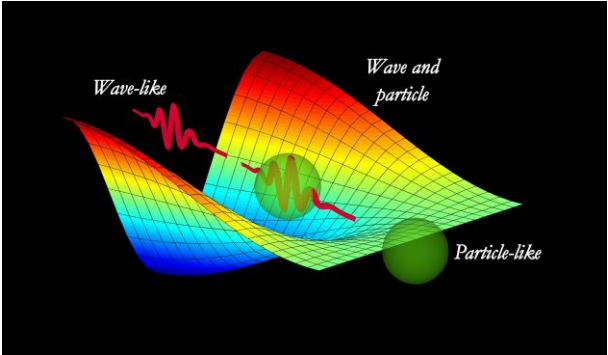
What We Are NOT Doing

- Not designing or trying to understand the inner workings of a quantum computer.
- Not learning how to do the fabricating quantum chips.
- Not going deep into semiconductor device physics or quantum physics.

What We WILL Learn

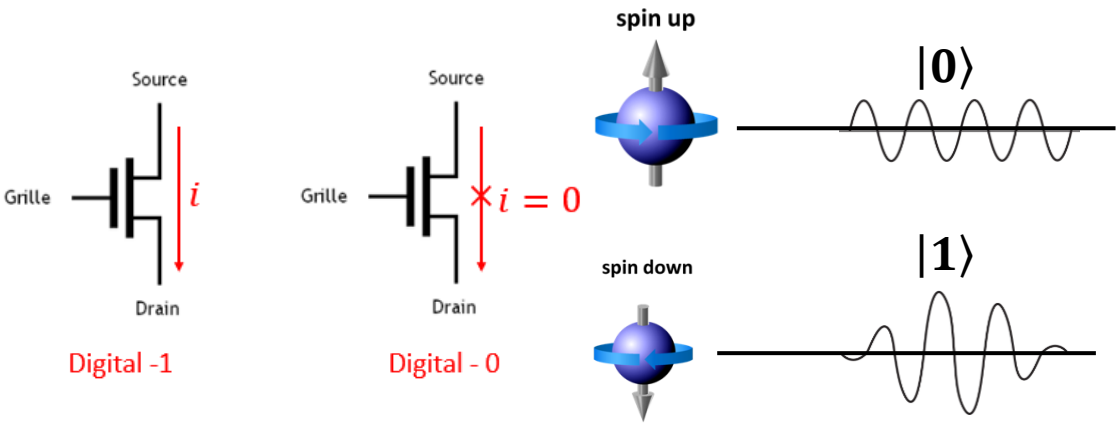
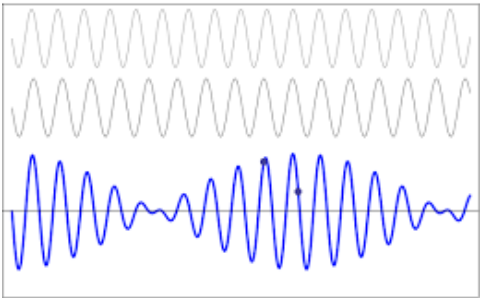
- How qubits are represented mathematically.
- How quantum gates manipulate amplitudes and phases.
- How circuits create interference and entanglement.
- How algorithms use these effects for computation.

Recap from L1



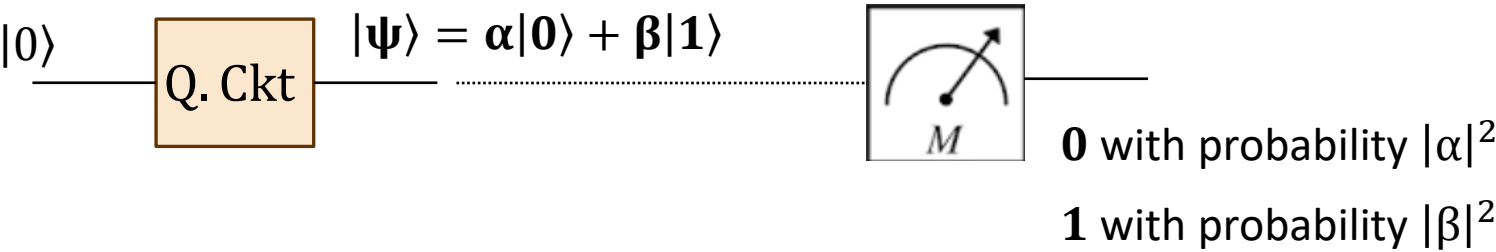
Quantum entities like electrons, which we know are particles, can also behave like a wave under special circumstances

Since they can behave like a wave, we can design experiments such that one quantum state can superpose on another quantum state



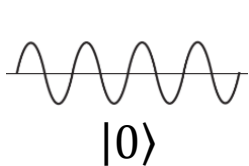
$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where $\{\alpha, \beta\} \in \mathbb{C}$

The wave nature allows us to write a composite quantum state that is a linear combination of the basis states, that is a quantum - 0 or $|0\rangle$ and quantum -1 or $|1\rangle$

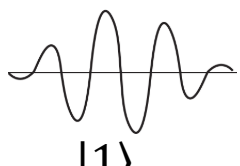


You can never measure a superposition of quantum states. You can create a superposition, you can do computation with it, but you can never measure a superposition

Motivation: The Magic is in Superposition



$|0\rangle$



$|1\rangle$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \text{ where } \{\alpha, \beta\} \in \mathbb{C}$$

1 qubit $\rightarrow$ 2 numbers $\rightarrow$ 2 bits of information

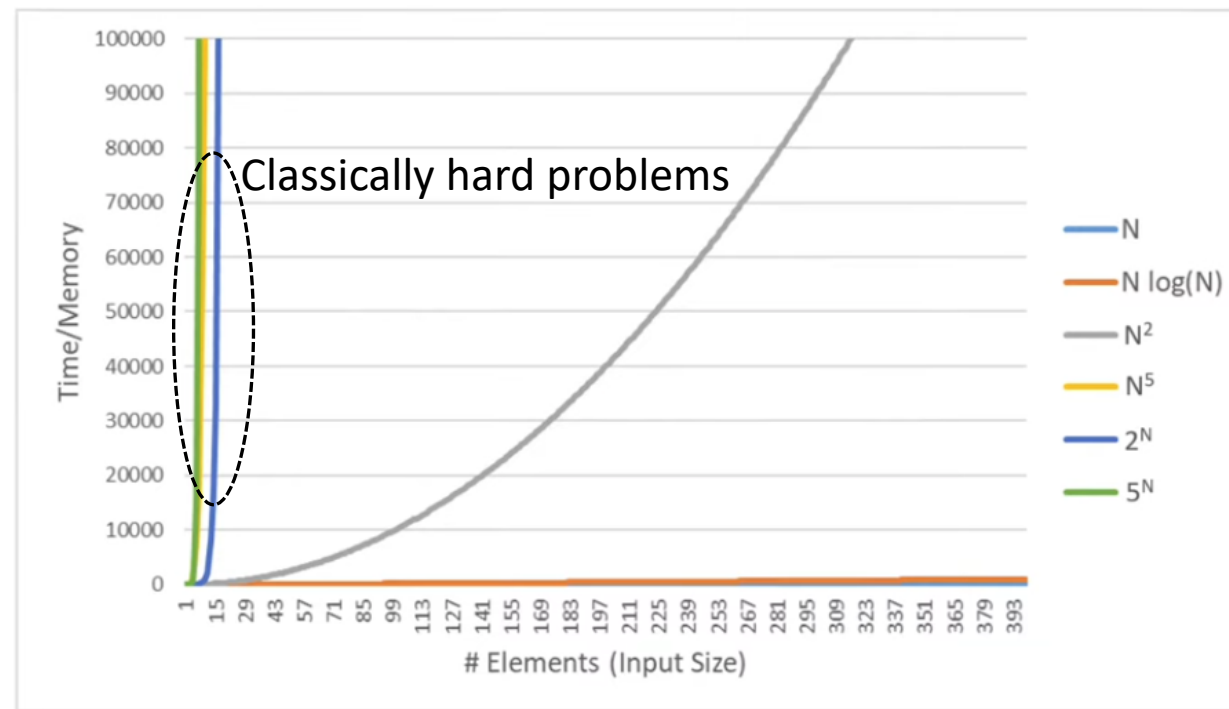
1 classical bit $\rightarrow$ 1 bit of information

$$|\psi\rangle = \alpha|0\rangle|0\rangle + \beta|0\rangle|1\rangle + \gamma|1\rangle|0\rangle + \delta|1\rangle|1\rangle, \{\alpha, \beta, \gamma, \delta\}$$

2 qubits prepared in a state of superposition
 $\rightarrow$ 4 numbers $\rightarrow$ 4 bits of information

n qubits prepared in a state of superposition
 $\rightarrow 2^n$ bits of information

An n-qubit quantum state is described by 2^n complex amplitudes (unlike n classical bits which store only n bits), meaning its mathematical description grows exponentially with the number of qubits.



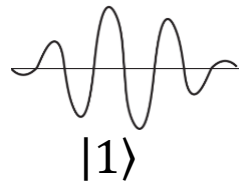
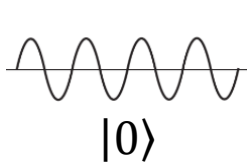
Drug Discovery & Molecular Simulation: Accurately simulating complex molecules (proteins, catalysts, battery materials).

Large-Scale Optimization (Logistics & Supply Chains): Optimizing airline schedules, delivery routes, warehouse planning.

Financial Modeling: Portfolio optimization with thousands of correlated assets.

Climate & Material Science Simulation: Modeling chemical reactions in carbon capture.

Understanding a Qubit



$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \text{ where } \{\alpha, \beta\} \in \mathbb{C}$$

$|\mathbf{x}\rangle$ called the “ket $\mathbf{x}$ ” is the state of a quantum entity (say a qubit) which is described by a unit column vector in a complex vector space (more generally called a Hilbert space): $|\mathbf{x}\rangle \in \mathbb{H}$

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \frac{i}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$|\psi\rangle = \frac{i}{2} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \frac{\sqrt{3}}{2} e^{i\pi/3} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \end{bmatrix} + \frac{i}{\sqrt{2}} \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

- So even though complex numbers may not feel “real” in everyday life, they are the natural language of waves. Since qubits behave like waves, complex numbers become essential.
- Think of waves in water video from L1. Two waves can add or cancel depending not just on their height, but on their phase (where they are in their cycle). Complex numbers are a mathematical way of keeping track of both: Amplitude (how big) as well as Phase (where in the cycle)
- A complex number is like an arrow in a plane:
 - Its length tells us the strength (amplitude).
 - Its angle tells us the phase.
 - This is exactly what we need to describe quantum states.
- In quantum computing, interference is everything. When we combine quantum states, we want some possibilities to reinforce and others to cancel out. Complex numbers allow this constructive and destructive interference to happen mathematically.

Detour: Complex Numbers

Some equations have no real solutions: $x^2 + 1 = 0 \rightarrow x = \pm i$ where $i^2 = -1$ is called an imaginary number

$4i, -i, 6i, -13i \dots$ are all examples of imaginary numbers

When we combine an imaginary number with a real number, we get a complex number

Complex Number: $z = a + ib$ $z \in \mathbb{C}$ but $\{a, b\} \in \mathbb{R}$

$\text{Re}(z) = a$ and $\text{Im}(z) = b$

Complex Conjugate: $z^* = a - ib$

Conjugate rules: $(z^*)^* = z$; $(z_1 z_2)^* = z_1^* z_2^*$; $(z_1 + z_2)^* = z_1^* + z_2^*$

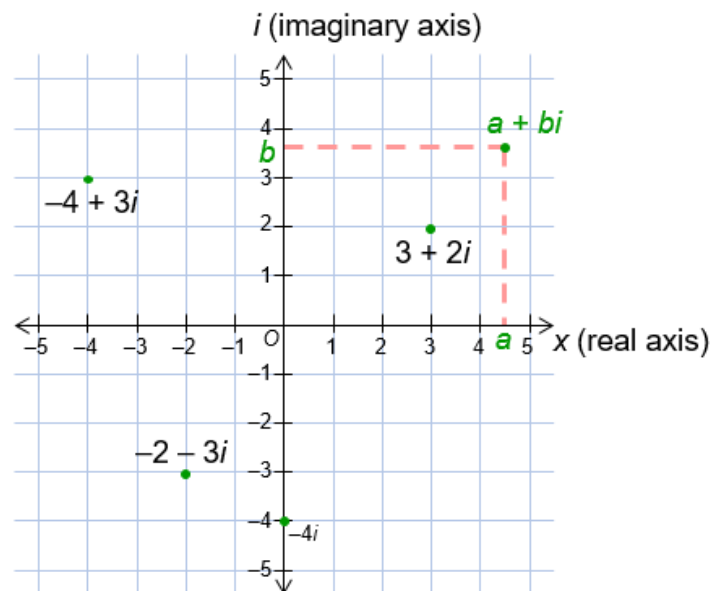
$$|z|^2 = zz^* = z^*z = a^2 + b^2$$

$$\text{Modulus: } |z| = \sqrt{a^2 + b^2}$$

Modulus rules: $|z_1 z_2| = |z_1| |z_2|$; $|z_1 + z_2| \neq |z_1| + |z_2|$

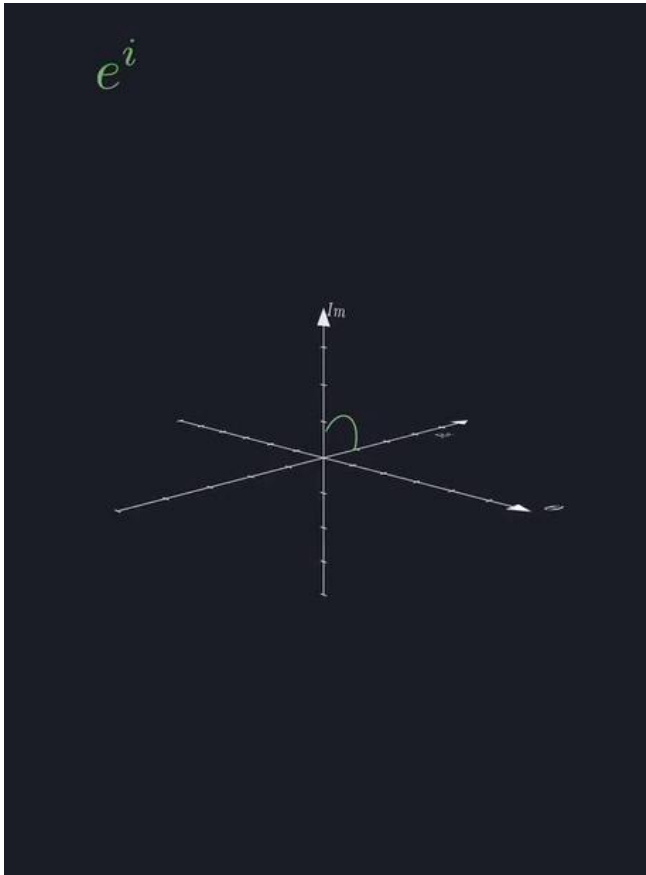
A modulus measures the length of a complex number.

Argand Diagram

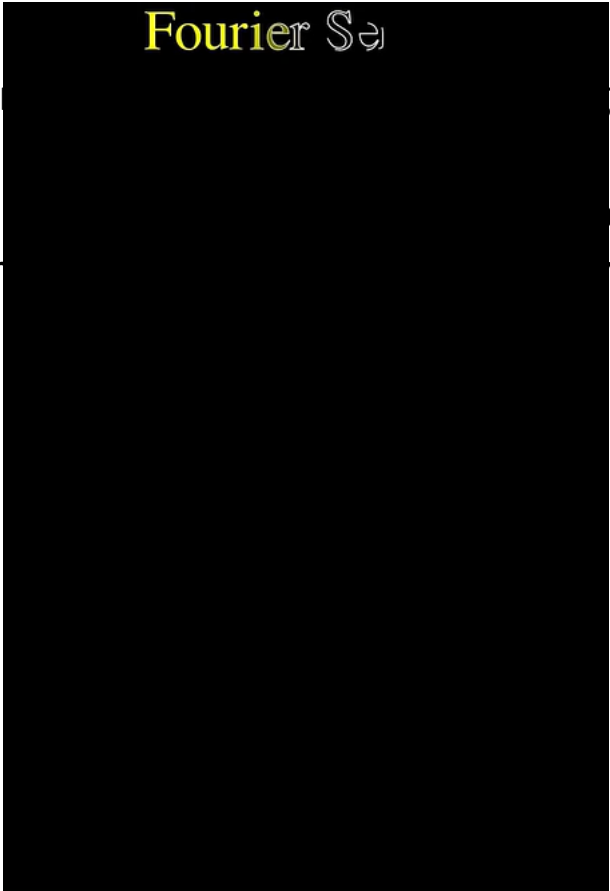
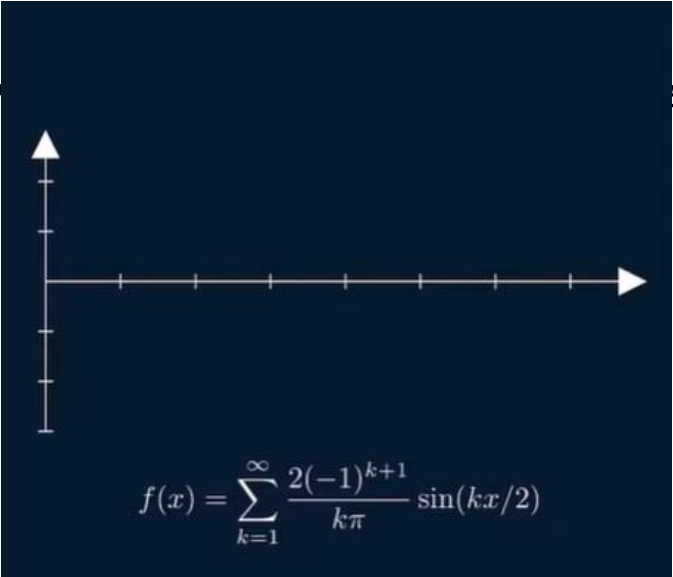


Detour: Complex Numbers

Oscillations and waves are easier with complex numbers: $e^{i\theta} = \cos\theta + i\sin\theta$ (Eulers Formula)



- We often write waves using sine and cosine because they are the simplest oscillating functions which are smooth, repeating, and mathematically clean.
- A deep mathematical result called Fourier’s theorem says that any reasonable wave (any repeating signal) can be written as a sum of sine and cosine waves of different frequencies.
- That is why sine and cosine are so important as building blocks of all



exponentials
including quantum

Qubit as a Vector in Complex Hilber Space

$$\theta = 0^\circ: e^{i0} = \cos 0 + i \sin 0 = 1$$

$$\theta = \frac{\pi}{2}: e^{i(\pi/2)} = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = i$$

$$\theta = \pi: e^{i\pi} = \cos \pi + i \sin \pi = -1 + 0 = -1$$

$$\theta = \frac{\pi}{4}: e^{i(\pi/4)} = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$$

$z = e^{i\theta}$ be a complex number. So, $z = \cos \theta + i \sin \theta$

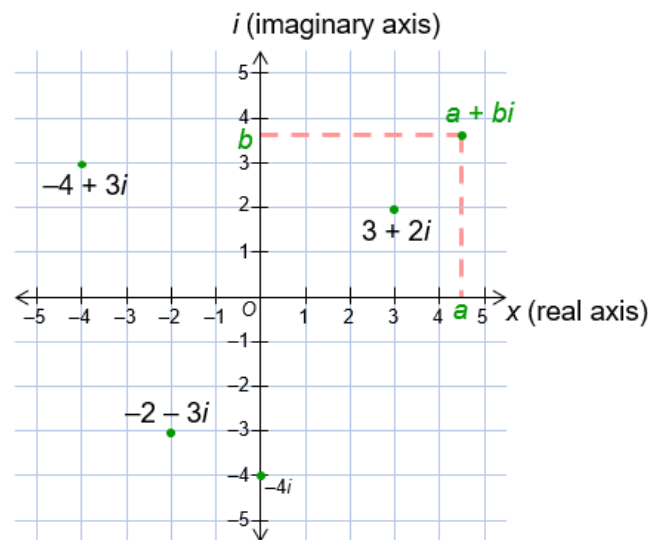
$\text{Re}(z) = a = \cos \theta$ and $\text{Im}(z) = b = \sin \theta$

Modulus: $|z| = \sqrt{a^2 + b^2} = \sqrt{\cos^2 \theta + \sin^2 \theta} = 1$

Any complex number can be expressed as $z = re^{i\theta}$

- $r = |z|$ (modulus — the length)
- $\theta = \arg(z)$ (the angle with the x-axis) = $\tan^{-1}(b/a)$

Argand Diagram



$$z = 3 + 2i \quad : a = 3; b = 2$$

$$\theta = \tan^{-1}(b/a) = \tan^{-1}(2/3) = 0.588 \text{ rad}$$

$$r = |z| = \sqrt{3^2 + 2^2} = \sqrt{13}$$

$$z = \sqrt{13}e^{i(0.588 \text{ rad})}$$

$$z = \sqrt{13}e^{i(34^\circ)}$$

$$z = -4 + 3i: a = -4; b = 3$$

$$\theta = \tan^{-1}(b/a) = \tan^{-1}(-3/4) = \pi - \tan^{-1}(3/4) = \pi - 0.644 \text{ rad} = 3.14 \text{ rad} - 0.644 \text{ rad} = 2.49 \text{ rad}$$

$$r = |z| = \sqrt{(-4)^2 + 3^2} = \sqrt{25} = 5$$

$$z = 5e^{i(2.49 \text{ rad})}$$

$$z = \sqrt{13}e^{i(143^\circ)} = \sqrt{13}e^{i(180-37^\circ)}$$

Qubit as a Vector in Complex Hilber Space

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \text{ where } \{\alpha, \beta\} \in \mathbb{C}$$

Recap from Linear Algebra Course:

- If $|v\rangle \in \mathbb{H}$ and α be any complex number then $\alpha|v\rangle$ also $\in \mathbb{H}$
- If $|v_1\rangle, |v_2\rangle \in \mathbb{H}$ then $\alpha|v_1\rangle + \beta|v_2\rangle$ also $\in \mathbb{H}$

$$\text{For } |v\rangle = \begin{bmatrix} v_1 \\ \vdots \\ v_d \end{bmatrix}_{d \times 1} \text{ where } v_1 \dots v_d \in \mathbb{C} \text{ be a ket-vector or } \mathbf{ket-v}$$

There is a corresponding row-vector or bra vector or **bra-v** such that $\langle v| = |v\rangle^\dagger = [v_1^* \dots v_d^*]_{1 \times d}$ where $v_1 \dots v_d \in \mathbb{C}$

$$\text{If } |v\rangle = \begin{bmatrix} 1 + i \\ 2 \end{bmatrix} \text{ then } \langle v| = [1 - i \quad 2].$$

$$\langle v|v\rangle = [1 - i \quad 2] \begin{bmatrix} 1 + i \\ 2 \end{bmatrix} = (1 - i^2) + 4 = 6$$

$\langle v|v\rangle$ is a bra-vector times a ket-vector. This notation was introduced by British physicist Paul Dirac. It is often called the **bra-ket** notation by computing community and as the Dirac notation by the Physics community

$$\text{Norm: } \|v\| = \sqrt{\langle v|v\rangle} = \sqrt{6}$$

A norm measures the length of a vector in $\mathbb{H}$

The vector $|v\rangle$ has a length of $\sqrt{6}$ which is not 1. This means the vector is not normalized. In quantum computing, we would often need vectors that are normalized

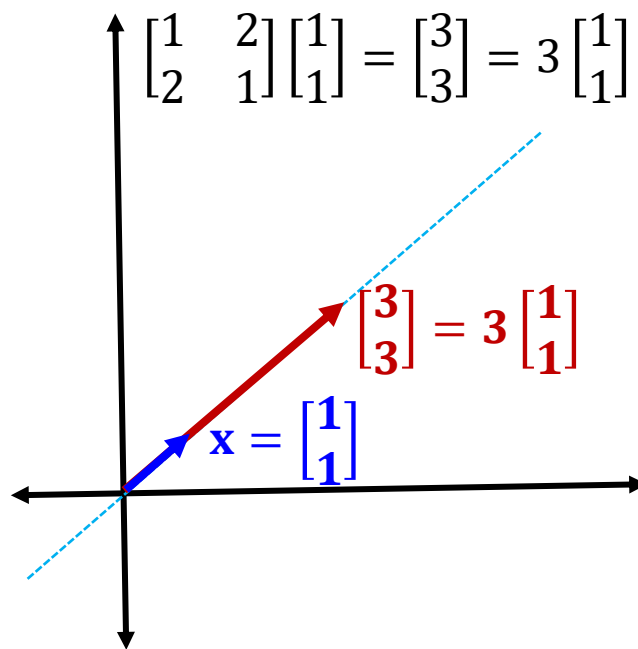
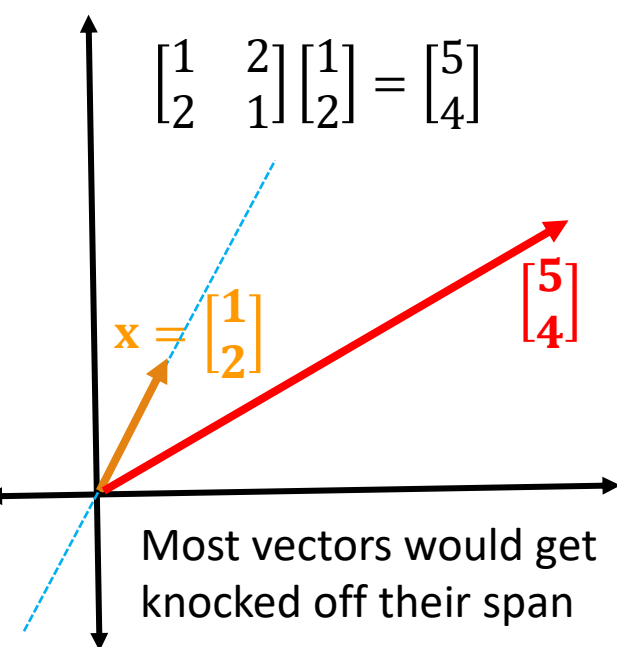
$$|v\rangle_{\text{norm}} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 + i \\ 2 \end{bmatrix}$$

Concepts from Linear Algebra

$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ and $x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, then $Ax = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ represents scaling (stretching/shrinking) a vector without changing the direction

$R = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}$ and $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, then $Rx = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ represents rotating a vector without changing its length

$M = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and $x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, then $Mx = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ represents flipping a vector without changing its length



- We are often interested in some special vectors, wherein, a matrix 'A' acting on a vector 'x' generates a new vector that is parallel (or antiparallel) to 'x'. The vector can get stretched or squeezed, but it stays on the span
- These "special" vectors are the **EIGEN VECTORS** associated with the matrix 'A'
- Mathematically, $Ax = \lambda x$ ('A' acting on 'x' generates a new vector, which is some multiple ' λ ' times the original vector 'x'). The multiple λ is an **EIGEN VALUE**

Concepts from Linear Algebra

$A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$. Find the eigen values and normalized eigen vectors?

$$|A - \lambda I| = 0 \Rightarrow \begin{bmatrix} 5 - \lambda & 4 \\ 4 & 5 - \lambda \end{bmatrix} = 0 \quad \lambda^2 - 10\lambda + 9 = 0 \Rightarrow \lambda_1 = 9 \text{ \& } \lambda_2 = 1 \text{ (Note, } \lambda_1 + \lambda_2 = \text{Trace of } [A] \text{ \& } \lambda_1 \lambda_2 = |A|)$$

To find Eigen vectors corresponding to Eigen value λ_1 : $\begin{bmatrix} 5 - \lambda_1 & 4 \\ 4 & 5 - \lambda_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} -4 & 4 \\ 4 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ or } c \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ will work. Lets chose with } \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

To find Eigen vectors corresponding to Eigen value λ_2 : $\begin{bmatrix} 5 - \lambda_2 & 4 \\ 4 & 5 - \lambda_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \text{ or } c \begin{bmatrix} -1 \\ 1 \end{bmatrix} \text{ will work. Lets chose with } \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Normalized eigen vectors would be $\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$ corresponding to $\lambda_1 = 9$ and $\begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$ corresponding to $\lambda_2 = 1$

Concepts from Linear Algebra

Hermitian Matrices: Hermitian matrices are those that are equal to their conjugate transpose ($H = H^\dagger$).

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}. \text{ Is it Hermitian?} \quad A^* = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \quad A^\dagger = (A^*)^T = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \quad \text{It is Hermitian}$$

$$A = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}. \text{ Is it Hermitian?} \quad A^\dagger = (A^*)^T = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} \neq A \quad \text{It is not Hermitian}$$

$$A = \begin{bmatrix} 2 & 1+i & -2i \\ 1-i & 3 & 4 \\ 2i & 4 & -1 \end{bmatrix}. \text{ Is it Hermitian?} \quad A^* = \begin{bmatrix} 2 & 1-i & +2i \\ 1+i & 3 & 4 \\ -2i & 4 & -1 \end{bmatrix} \quad A^\dagger = (A^*)^T = \begin{bmatrix} 2 & 1+i & -2i \\ 1-i & 3 & 4 \\ +2i & 4 & -1 \end{bmatrix} = A$$

Yes, Hermitian

$$A = \begin{bmatrix} 1 & 2+i & 3 \\ 2-i & 0 & 4i \\ 3 & 5 & -2 \end{bmatrix}. \text{ Is it Hermitian?} \quad A^\dagger = (A^*)^T = \begin{bmatrix} 1 & 2-i & 3 \\ 2+i & 0 & -4i \\ 3 & 5 & -2 \end{bmatrix}^T = \begin{bmatrix} 1 & 2+i & 3 \\ 2-i & 0 & 5 \\ 3 & -4i & -2 \end{bmatrix} \neq A$$

It is not Hermitian

Concepts from Linear Algebra

In quantum mechanics, Hermitian matrices represent measurements. Their eigenvalues are always real numbers, because the outcome of any physical measurement must be real. Complex numbers are mathematical tools that help us describe quantum states and interference, but the final result of an experiment can never be imaginary or complex. Therefore, quantum measurements are always represented by Hermitian matrices, ensuring real measurable outcomes.

$$H = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}. \text{ Is it Hermitian?} \quad H^* = \begin{bmatrix} 0 & -i \\ +i & 0 \end{bmatrix} \quad H^\dagger = (H^*)^T = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix} = H$$

$$|H - \lambda I| = 0 \Rightarrow \begin{bmatrix} -\lambda & i \\ -i & -\lambda \end{bmatrix} = 0 \quad \lambda^2 - 1 = 0 \Rightarrow \lambda_1 = 1 \text{ \& } \lambda_2 = -1 \text{ (Real eigen values)}$$

(Note, $\lambda_1 + \lambda_2 = \text{Trace of } [H] = \text{Sum of diagonal elements}$ & $\lambda_1 \lambda_2 = |H|$)

$$H = \begin{bmatrix} 2 & 1+i \\ 1-i & 3 \end{bmatrix}. \text{ Is it Hermitian?} \quad H^* = \begin{bmatrix} 2 & 1-i \\ 1+i & 3 \end{bmatrix} \quad H^\dagger = (H^*)^T = \begin{bmatrix} 2 & 1+i \\ 1-i & 3 \end{bmatrix} = H$$

$$|H - \lambda I| = 0 \Rightarrow \begin{bmatrix} 2-\lambda & 1+i \\ 1-i & 3-\lambda \end{bmatrix} = 0 \quad \lambda^2 - 5\lambda + 4 = 0 \Rightarrow \lambda_1 = 1 \text{ \& } \lambda_2 = 4 \text{ (Real eigen values)}$$

Concepts from Linear Algebra

Unitary Matrices: Unitary matrices have a key property $U^\dagger U = I = U U^\dagger$. Unitary matrices represent allowed quantum gates or quantum operations. They change quantum states without destroying information.

$$U = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \text{ Is this Unitary?}$$

$$U^\dagger = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}. \text{ So, } U^\dagger U = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$U = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \text{ Is this Unitary?}$$

$$U^\dagger = \begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix}. \text{ So, } U^\dagger U = \begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -i^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix} \text{ Is this Unitary?}$$

$$U^\dagger = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix} \quad U^\dagger U = \frac{1}{2} \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Since $U^\dagger U = I$, note that $U^{-1} = U^\dagger$

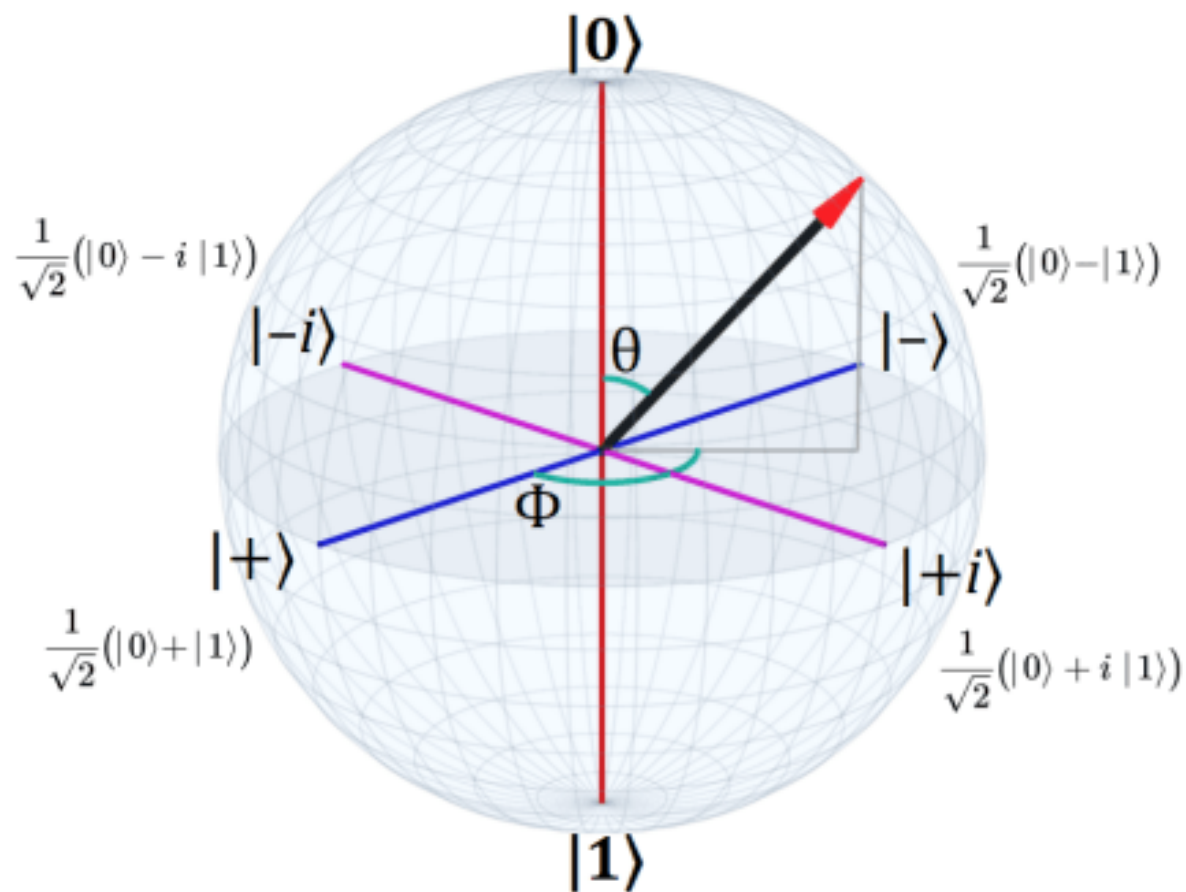
If $U = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix}$ be a unitary operation,

then the reverse (inverse) operation $U^{-1} = U^\dagger = \begin{bmatrix} 1 & 0 \\ 0 & e^{-i\theta} \end{bmatrix}$

This means, if $U|\psi\rangle = |\varphi\rangle$ transforms the initial state vector $|\psi\rangle$ to the final state vector $|\varphi\rangle$ through U , then the operation can be reversed or un-done by $U^\dagger|\varphi\rangle = |\psi\rangle$ through $U^\dagger$.

Except for measurement, which destroys superposition and collapses the quantum state, all quantum gate operations are reversible because they are represented by unitary transformations in Hilbert space. This is a fundamental contrast with classical computing, where most logic gates (like AND, OR, NAND) are irreversible and lose information.

Bloch Sphere Representation of Qubits



- The Bloch sphere represents all possible pure states of a single qubit. Every point on its surface corresponds to a valid quantum state.
- All points on the equator are equal superpositions of $|0\rangle$ and $|1\rangle$. If measured in the Z-basis, they give 0 with probability 1/2 and 1 with probability 1/2.
- What fundamentally distinguishes states like $|+\rangle$, $|+i\rangle$, $|-\rangle$, $|-\rangle$ is their relative phase. Even though their measurement probabilities are identical, they evolve differently under quantum gates and interfere differently in quantum circuits.
- From a computation point of view, amplitudes determine what you measure, but phases determine how states interfere and transform and that phase information is what gives quantum computing its advantage.

Basis Transformation

A quantum state does not change when you change the basis. Only our description of it changes.

Standard Basis or Z-basis $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ in 2D (This is what we will mostly deal with in this course)

$$\langle 0|0\rangle = \langle 1|1\rangle = 1 \quad \langle 0|1\rangle = \langle 1|0\rangle = 0$$

Standard basis is an orthonormal basis

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$\| |+\rangle \| = \sqrt{\langle +|+ \rangle} = \left[\left\{ \frac{\langle 0| + \langle 1|}{\sqrt{2}} \right\} \left\{ \frac{|0\rangle + |1\rangle}{\sqrt{2}} \right\} \right]^{1/2}$$

$$\left[\frac{\langle 0|0\rangle + \langle 0|1\rangle + \langle 1|0\rangle + \langle 1|1\rangle}{2} \right]^{1/2} = 1$$

is called the Hadamard basis or X-basis

$$\text{Check: } \| |-\rangle \| = \sqrt{\langle -|-\rangle} = 1$$

$$\langle +|-\rangle = \left\{ \frac{\langle 0| + \langle 1|}{\sqrt{2}} \right\} \left\{ \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right\} = \frac{\langle 0|0\rangle - \langle 0|1\rangle + \langle 1|0\rangle - \langle 1|1\rangle}{2} = 0$$

Hadamard basis is also an orthonormal basis

Basis Transformation

If we have an orthonormal basis, say $\{|\alpha\rangle, |\beta\rangle\}$, and let's say we have a qubit in state $|\psi\rangle$ which is some other basis other than $\{|\alpha\rangle, |\beta\rangle\}$.

The way to express the qubit state in $\{|\alpha\rangle, |\beta\rangle\}$ basis is $|\psi\rangle_{\alpha,\beta} = \langle\alpha|\psi\rangle|\alpha\rangle + \langle\beta|\psi\rangle|\beta\rangle$.

Note, $\langle\alpha|\psi\rangle$ will determine the probability of measuring $|\alpha\rangle$, that is $|\langle\alpha|\psi\rangle|^2$ and $\langle\beta|\psi\rangle$ will determine the probability of measuring $|\beta\rangle$, that is $|\langle\beta|\psi\rangle|^2$

$|\psi\rangle = |0\rangle$ is already in the standard basis

Prob. of measuring 0 is 1

Prob. of measuring 1 is 0

Express $|\psi\rangle$ in the Hadamard basis

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

We know that $|\psi\rangle_{\alpha,\beta} = \langle\alpha|\psi\rangle|\alpha\rangle + \langle\beta|\psi\rangle|\beta\rangle$

Therefore, $|\psi\rangle_{+,-} = \langle+|0\rangle|+\rangle + \langle-|0\rangle|-\rangle$

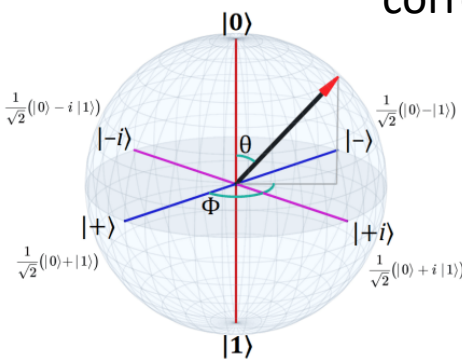
To evaluate $\langle+|0\rangle$, let's first express the bra-vectors of the corresponding et-vectors

$$\langle+| = \frac{\langle 0| + \langle 1|}{\sqrt{2}} \quad \langle-| = \frac{\langle 0| - \langle 1|}{\sqrt{2}}$$

$$\begin{aligned} |\psi\rangle_{+,-} &= \left\{ \frac{\langle 0| + \langle 1|}{\sqrt{2}} |0\rangle \right\} |+\rangle + \left\{ \frac{\langle 0| - \langle 1|}{\sqrt{2}} |0\rangle \right\} |-\rangle \\ &= \left\{ \frac{\langle 0|0\rangle + \langle 1|0\rangle}{\sqrt{2}} \right\} |+\rangle + \left\{ \frac{\langle 0|0\rangle - \langle 1|0\rangle}{\sqrt{2}} \right\} |-\rangle = \frac{1}{\sqrt{2}} |+\rangle + \frac{1}{\sqrt{2}} |-\rangle \end{aligned}$$

Prob. of measuring $|+\rangle$ is 1/2

Prob. of measuring $|-\rangle$ is 1/2



Basis Transformation

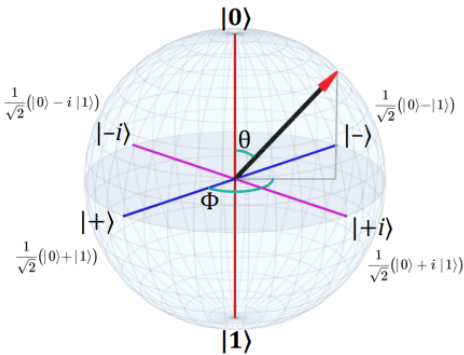
The way to express the qubit state in $\{|\alpha\rangle, |\beta\rangle\}$ basis is $|\psi\rangle_{\alpha,\beta} = \langle\alpha|\psi\rangle|\alpha\rangle + \langle\beta|\psi\rangle|\beta\rangle$.

Note, $\langle\alpha|\psi\rangle$ will determine the probability of measuring $|\alpha\rangle$, that is $|\langle\alpha|\psi\rangle|^2$ and $\langle\beta|\psi\rangle$ will determine the probability of measuring $|\beta\rangle$, that is $|\langle\beta|\psi\rangle|^2$

$$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$$

Prob. of measuring 0 is = 3/4

Prob. of measuring 1 is 1/4



$$\begin{aligned} |\psi\rangle_{+,-} &= \left\{ \frac{\langle 0| + \langle 1|}{\sqrt{2}} |\psi\rangle \right\} |+\rangle + \left\{ \frac{\langle 0| - \langle 1|}{\sqrt{2}} |\psi\rangle \right\} |-\rangle \\ &= \left\{ \frac{\langle 0| + \langle 1|}{\sqrt{2}} \left(\frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle \right) \right\} |+\rangle + \left\{ \frac{\langle 0| - \langle 1|}{\sqrt{2}} \left(\frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle \right) \right\} |-\rangle \end{aligned}$$

Express $|\psi\rangle$ in the Hadamard basis

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

is the Hadamard basis in 2D

$$|\psi\rangle_{+,-} = \langle +|\psi\rangle|+\rangle + \langle -|\psi\rangle|-\rangle$$

$$\langle +| = \frac{\langle 0| + \langle 1|}{\sqrt{2}} \quad \langle -| = \frac{\langle 0| - \langle 1|}{\sqrt{2}}$$

$$\begin{aligned} &= \left\{ \frac{\sqrt{3}}{2\sqrt{2}} \langle 0|0\rangle + \frac{1}{2\sqrt{2}} \langle 0|1\rangle + \frac{\sqrt{3}}{2\sqrt{2}} \langle 1|0\rangle + \frac{1}{2\sqrt{2}} \langle 1|1\rangle \right\} |+\rangle \\ &+ \left\{ \frac{\sqrt{3}}{2\sqrt{2}} \langle 0|0\rangle + \frac{1}{2\sqrt{2}} \langle 0|1\rangle - \frac{\sqrt{3}}{2\sqrt{2}} \langle 1|0\rangle - \frac{1}{2\sqrt{2}} \langle 1|1\rangle \right\} |-\rangle \end{aligned}$$

$$|\psi\rangle_{+,-} = \frac{\sqrt{3} + 1}{2\sqrt{2}} |+\rangle + \frac{\sqrt{3} - 1}{2\sqrt{2}} |-\rangle$$

$$P(+)=\left|\frac{\sqrt{3}+1}{2\sqrt{2}}\right|^2=0.933 \quad P(-)=\left|\frac{\sqrt{3}-1}{2\sqrt{2}}\right|^2=0.067$$

Basis Transformation

$|\psi\rangle = \left(\frac{1}{\sqrt{6}} - \frac{1}{\sqrt{3}}\right)|+\rangle + \left(\frac{1}{\sqrt{6}} + \frac{1}{\sqrt{3}}\right)|-\rangle$ is in the Hadamard basis

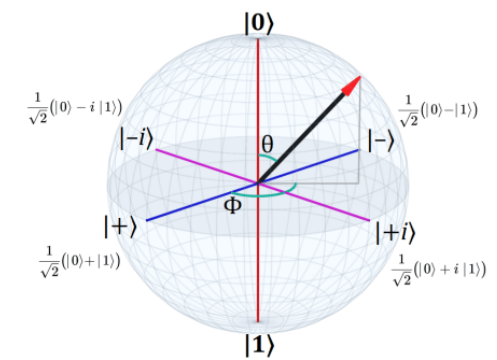
$$P(+)=\left|\left(\frac{1}{\sqrt{6}}-\frac{1}{\sqrt{3}}\right)\right|^2=\frac{3-2\sqrt{2}}{6}=0.0287$$

$$P(-)=\left|\left(\frac{1}{\sqrt{6}}+\frac{1}{\sqrt{3}}\right)\right|^2=\frac{3+2\sqrt{2}}{6}=0.9713$$

Express $|\psi\rangle$ in the standard basis and find $P(1)$

$$|\psi\rangle_{0,1}=\langle 0|\psi\rangle|0\rangle+\langle 1|\psi\rangle|1\rangle$$

$$\begin{aligned}\langle 0|\psi\rangle &= \langle 0|\left\{\left(\frac{1}{\sqrt{6}}-\frac{1}{\sqrt{3}}\right)\frac{|0\rangle+|1\rangle}{\sqrt{2}}+\left(\frac{1}{\sqrt{6}}+\frac{1}{\sqrt{3}}\right)\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right\}\\&= \left\{\left(\frac{1}{\sqrt{6}}-\frac{1}{\sqrt{3}}\right)\frac{\langle 0|0\rangle+\langle 0|1\rangle}{\sqrt{2}}+\left(\frac{1}{\sqrt{6}}+\frac{1}{\sqrt{3}}\right)\frac{\langle 0|0\rangle-\langle 0|1\rangle}{\sqrt{2}}\right\}\\&= \frac{1}{\sqrt{2}}\left\{\left(\frac{1}{\sqrt{6}}-\frac{1}{\sqrt{3}}\right)+\left(\frac{1}{\sqrt{6}}+\frac{1}{\sqrt{3}}\right)\right\}=\frac{1}{\sqrt{3}}\end{aligned}$$



$$\begin{aligned}\langle 1|\psi\rangle &= \langle 1|\left\{\left(\frac{1}{\sqrt{6}}-\frac{1}{\sqrt{3}}\right)\frac{|0\rangle+|1\rangle}{\sqrt{2}}+\left(\frac{1}{\sqrt{6}}+\frac{1}{\sqrt{3}}\right)\frac{|0\rangle-|1\rangle}{\sqrt{2}}\right\}\\&= \left\{\left(\frac{1}{\sqrt{6}}-\frac{1}{\sqrt{3}}\right)\frac{\langle 1|0\rangle+\langle 1|1\rangle}{\sqrt{2}}+\left(\frac{1}{\sqrt{6}}+\frac{1}{\sqrt{3}}\right)\frac{\langle 1|0\rangle-\langle 1|1\rangle}{\sqrt{2}}\right\}\\&= \frac{1}{\sqrt{2}}\left\{\left(\frac{1}{\sqrt{6}}-\frac{1}{\sqrt{3}}\right)-\left(\frac{1}{\sqrt{6}}+\frac{1}{\sqrt{3}}\right)\right\}=-\sqrt{\frac{2}{3}}\end{aligned}$$

$$|\psi\rangle_{0,1}=\frac{1}{\sqrt{3}}|0\rangle-\sqrt{\frac{2}{3}}|1\rangle$$

$$P(1)=\left|-\sqrt{\frac{2}{3}}\right|^2=\frac{2}{3}$$

Basis Transformation

$$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle \text{ is in the standard basis}$$

Express $|\psi\rangle$ in the $-i/-i$ basis and find $P(+i)$

$$|+i\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}} \quad |-i\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}}$$

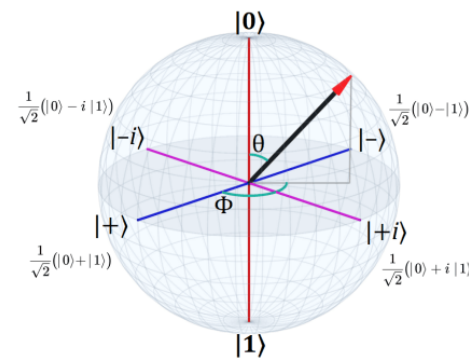
$$\langle +i| = \frac{\langle 0| - i\langle 1|}{\sqrt{2}} \quad \langle -i| = \frac{\langle 0| + i\langle 1|}{\sqrt{2}}$$

$$|\psi\rangle_{+i,-i} = \langle +i|\psi\rangle|+i\rangle + \langle -i|\psi\rangle|-i\rangle$$

$$\langle +i|\psi\rangle = \frac{\langle 0| - i\langle 1|}{\sqrt{2}} \left(\frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle \right)$$

$$= \frac{\sqrt{3}}{2\sqrt{2}}\langle 0|0\rangle + \frac{-i^2}{2\sqrt{2}}\langle 1|1\rangle$$

$$= \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

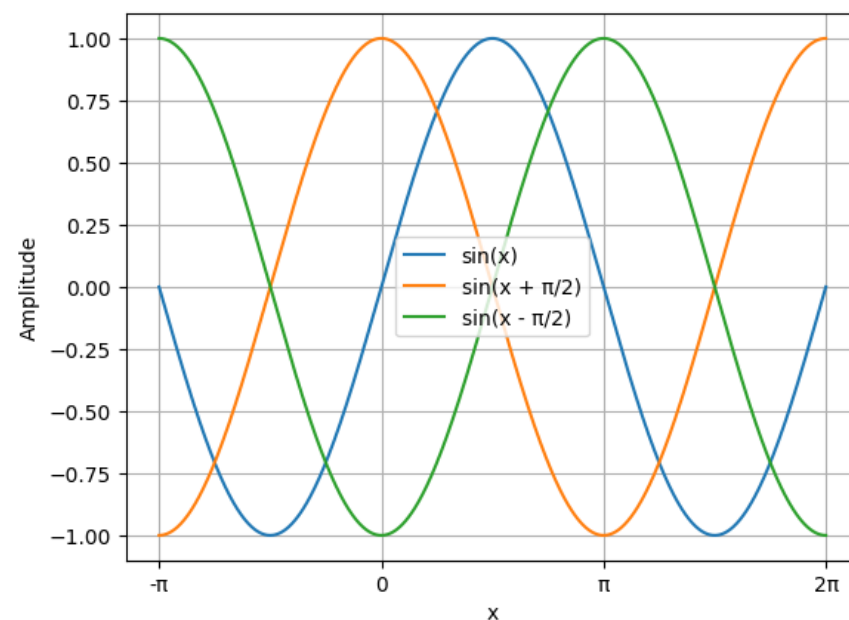
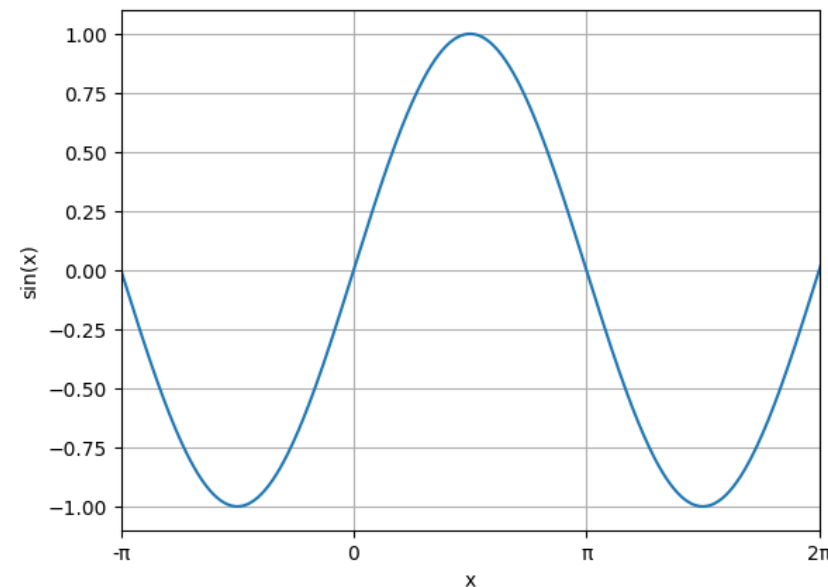
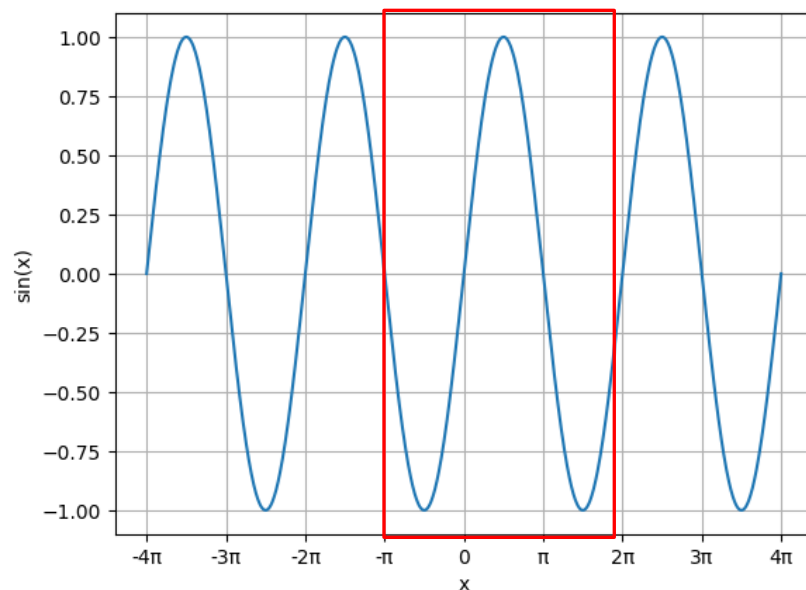


$$\begin{aligned} \langle -i|\psi\rangle &= \frac{\langle 0| + i\langle 1|}{\sqrt{2}} \left(\frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle \right) \\ &= \frac{\sqrt{3}}{2\sqrt{2}}\langle 0|0\rangle + \frac{i^2}{2\sqrt{2}}\langle 1|1\rangle \\ &= \frac{\sqrt{3} - 1}{2\sqrt{2}} \end{aligned}$$

$$|\psi\rangle_{+i,-i} = \frac{\sqrt{3} + 1}{2\sqrt{2}}|+i\rangle + \frac{\sqrt{3} - 1}{2\sqrt{2}}|-i\rangle$$

$$P(+i) = \left| \frac{\sqrt{3} + 1}{2\sqrt{2}} \right|^2 = 0.933$$

Global and Relative Phase of a Qubit



- A wave like $\sin(x)$ can be shifted left or right. This shift is called the phase of the wave. If we write a wave as $\sin(x \pm \theta)$, the number θ tells us how much the wave is shifted. The amplitude and frequency stay the same while only the phase changes.
- Using complex numbers, waves are written compactly as $e^{\pm i\theta}$. Here, θ represents the phase. Multiplying a wave by $e^{\pm i\theta}$ adds(+) /subtracts (−) a phase shift of θ . In other words, complex exponentials make adding phase mathematically simple.
- In quantum computing, qubit amplitudes are complex numbers. Their phase differences determine how quantum states interfere and interference is what enables quantum computation

Global and Relative Phase of a Qubit

If we have a qubit in state $|\psi\rangle$, then $e^{i\theta}|\psi\rangle$ represent the same qubit, but with a global phase shift of θ

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad \begin{array}{l} \text{Prob. of measuring } |0\rangle = 1/2 \\ \text{Prob. of measuring } |1\rangle = 1/2 \end{array}$$

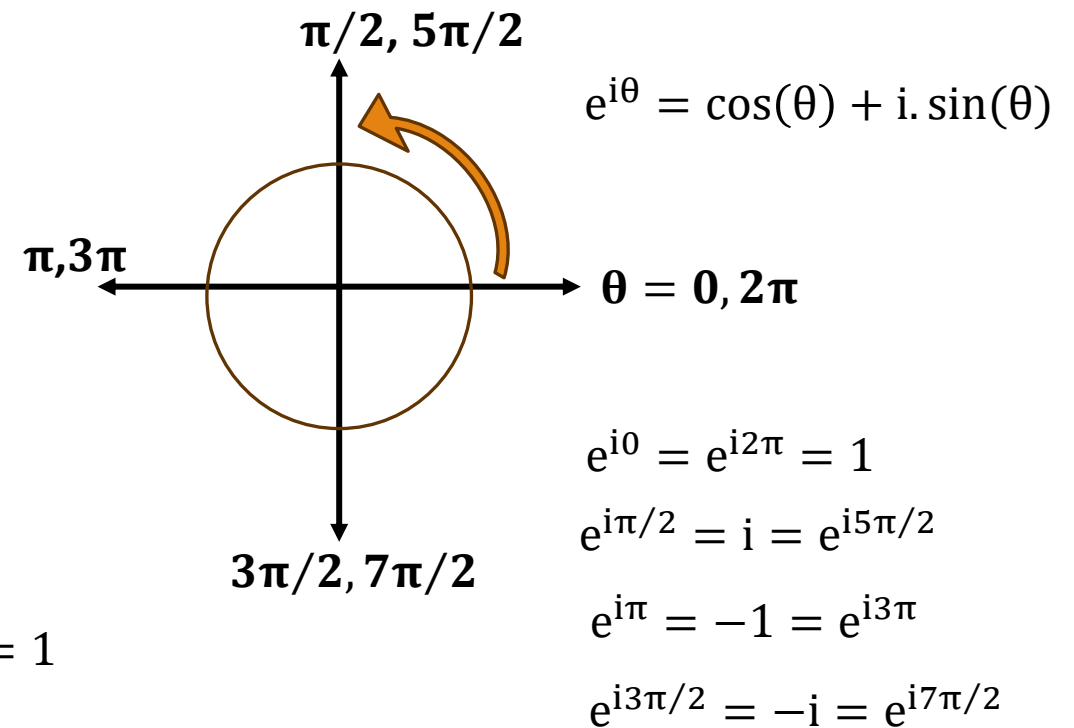
Adding a global phase of $\theta = \pi/2$: $e^{i\pi/2}|\psi\rangle = \frac{e^{i\pi/2}|0\rangle}{\sqrt{2}} + \frac{e^{i\pi/2}|1\rangle}{\sqrt{2}}$

$$\text{Prob. of measuring } |0\rangle = \left| \frac{e^{i\pi/2}}{\sqrt{2}} \right|^2 = 1/2 \quad \text{Recall: } |e^{i\theta}|^2 = e^{i\theta} \cdot e^{-i\theta} = 1$$

$$\text{Prob. of measuring } |1\rangle = \left| \frac{e^{i\pi/2}}{\sqrt{2}} \right|^2 = 1/2$$

Global phase is physically unobservable.

This is why quantum states are defined up to a global phase.



$|\psi\rangle$ and $-|\psi\rangle$ represent the same quantum state, but with a relative phase difference of π

$|\psi\rangle$ and $i|\psi\rangle$ represent the same quantum state, but with a relative phase difference of $\pi/2$

$|\psi\rangle$ and $-i|\psi\rangle$ represent the same quantum state, but with a relative phase difference of $3\pi/2$

Global and Relative Phase of a Qubit

$$|\psi\rangle = \frac{|0\rangle}{\sqrt{2}} + \frac{|1\rangle}{\sqrt{2}} \quad \text{When measured in the } \{|0\rangle, |1\rangle\} \text{ basis}$$

Prob. of measuring $|0\rangle = 1/2$

$$|\varphi\rangle = \frac{|0\rangle}{\sqrt{2}} - \frac{|1\rangle}{\sqrt{2}} \quad \text{Prob. of measuring } |1\rangle = 1/2$$

$|\varphi\rangle$ is the same as $|\psi\rangle$ except as phase of π (Note: $e^{i\pi} = -1$) has been added to $|1\rangle$

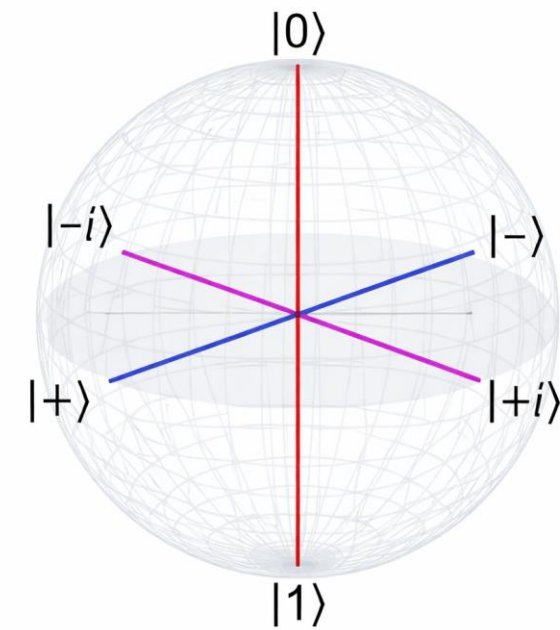
$|\psi\rangle$ and $|\varphi\rangle$ are undisguisable though measurements in the $\{|0\rangle, |1\rangle\}$ basis

But if we make a basis transformation from the $\{|0\rangle, |1\rangle\}$ basis to the $\{|+\rangle, |-\rangle\}$ basis

For $|\psi\rangle$: Prob. of measuring $|+\rangle = 1$
Prob. of measuring $|-\rangle = 0$

For $|\varphi\rangle$: Prob. of measuring $|+\rangle = 0$
Prob. of measuring $|-\rangle = 1$

$|\psi\rangle$ and $|\varphi\rangle$ are now distinguishable though measurements since the measurement was done in the appropriate basis



- **Global phase is unobservable:** A common phase factor does not change any measurement outcome.

- **Relative phase is physically meaningful:** Phase differences between components affect interference and results.

- **Phase becomes visible only in the right basis:** A basis change can convert phase information into probabilities.

- **Quantum algorithms work by manipulating phase, not probability:** Probabilities emerge only at the final measurement.

- **Basis transformations are a tool, not a change of reality:** The state is unchanged; what becomes measurable is not.

Quantum Gates (Always Represented by Unitary Transformations)

$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. This matrix is a Unitary matrix and represents the quantum version of the NOT gate. We call it the Pauli X-gate.

In the Bloch sphere representation, it denotes a Bit-flip of a qubit by π around the x-axis. Note $X^2 = I$

$X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$ $X|1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$ $|\psi\rangle \xrightarrow{\text{X}} |\varphi\rangle = X|\psi\rangle$

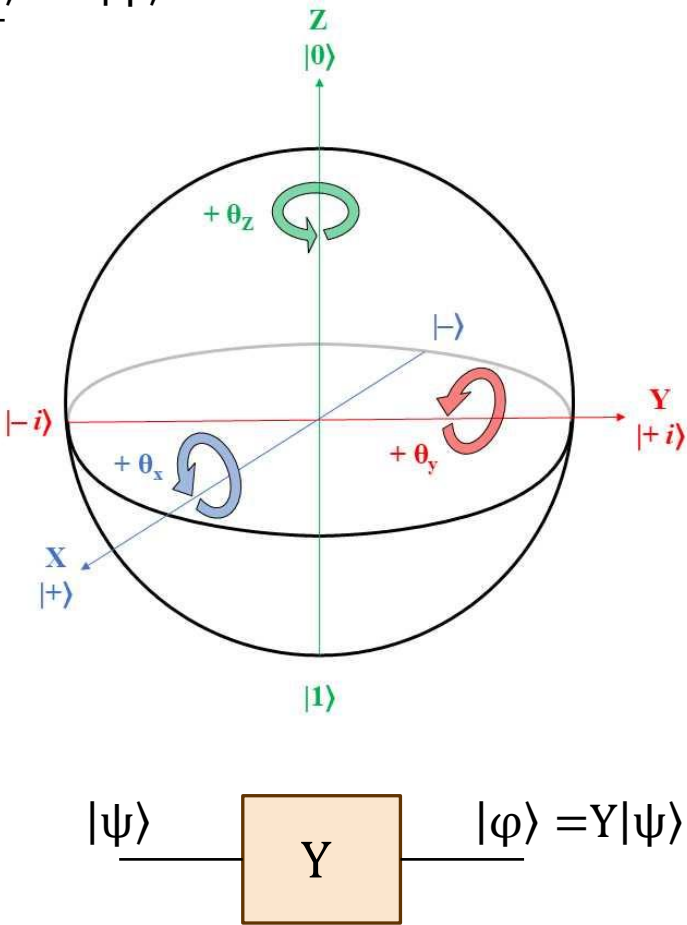
$X|\psi\rangle = X[\alpha|0\rangle + \beta|1\rangle] = \alpha X|0\rangle + \beta X|1\rangle = \beta|0\rangle + \alpha|1\rangle$

$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ is called the Pauli Z-gate. In the Bloch sphere representation, it denotes a Phase-flip of a qubit by π around the z-axis. $Z^2 = I$

$Z|0\rangle = |0\rangle; Z|1\rangle = -|1\rangle$ $|\psi\rangle \xrightarrow{\text{Z}} |\varphi\rangle = Z|\psi\rangle$

$Z|\psi\rangle = Z[\alpha|0\rangle + \beta|1\rangle] = \alpha Z|0\rangle + \beta Z|1\rangle = \alpha|0\rangle - \beta|1\rangle$

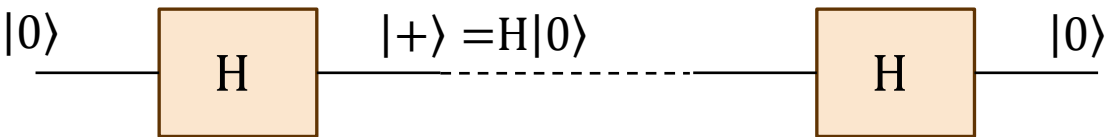
$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ is called the Pauli Y-gate. In the Bloch sphere representation, it denotes both Bit-flip and Phase-flip of a qubit by π around the y-axis. The Pauli X-, Y-, Z-gate and are single qubit gates. $Y^2 = I$



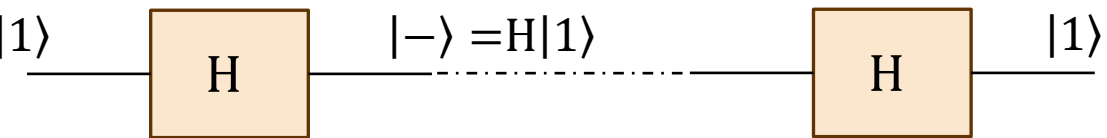
Quantum Gates (Always Represented by Unitary Transformations)

$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ Is called the Hadamard gate. It is one of the most extensively used single qubit gate in quantum circuits. It is used to create an equal superposition of $|0\rangle$ and $|1\rangle$

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle$$



$$H|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle$$



$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$H|\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} \alpha + \beta \\ \alpha - \beta \end{bmatrix}$$

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \sqrt{3}/2 \\ 1/2 \end{bmatrix} \quad H|\psi\rangle = \frac{1}{2\sqrt{2}} \begin{bmatrix} \sqrt{3} + 1 \\ \sqrt{3} - 1 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3} + 1}{2\sqrt{2}} \\ \frac{\sqrt{3} - 1}{2\sqrt{2}} \end{bmatrix}$$

$$H|+\rangle = H \left\{ \frac{|0\rangle + |1\rangle}{\sqrt{2}} \right\} = \frac{H|0\rangle}{\sqrt{2}} + \frac{H|1\rangle}{\sqrt{2}} = \frac{|0\rangle + |1\rangle}{\sqrt{2}\sqrt{2}} + \frac{|0\rangle - |1\rangle}{\sqrt{2}\sqrt{2}} = |0\rangle$$

- The qubit was already in **superposition**
- Hadamard **mixed the amplitudes**
- Probabilities changed due to **interference**

$H|-\rangle = |1\rangle$ This property of reversibility is not just unique to the H-gate. In fact, all quantum gates are reversible gates because they are unitary transformations.
In contrast, all classical gates are non-reversible (except the single input NOT gate).

Summary

Qubits & Complex Numbers: A qubit is a unit vector in complex Hilbert space: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. Complex numbers via $e^{i\theta} = \cos(\theta) + i \cdot \sin(\theta)$ let us track amplitude + phase, enabling interference, which gives quantum advantage.

Bloch Sphere & Phase

The Bloch sphere visualizes all pure single-qubit states.

- Amplitudes $\rightarrow$ measurement probabilities
- Relative phase $\rightarrow$ interference & evolution
- Global phase $\rightarrow$ physically unobservable

Bases & Information Extraction

Different bases (Z: $|0\rangle, |1\rangle$ or the X: $|+\rangle, |-\rangle$) reveal different information. Phase may be invisible in one basis but becomes measurable after a basis transformation.

Quantum Gates = Unitary Transformations

All quantum gates satisfy $U^\dagger U = I \rightarrow$ reversible operations. Measurement is the only non-unitary step (collapses superposition).

Key Gates

X: Bit flip; **Z:** Phase flip; **Y:** Bit + phase flip;

H (Hadamard): Creates superposition & changes basis — foundational for interference.